

Chemistry — Atoms, pH, Soaps, Enthalpy and Everyday Substances

Chapter F9.3 — Containing Three of the Ten 2023 Science Items

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NOTE · WHY THREE SECTIONS MATTER MORE THAN THE REST

The 2023 paper asked three chemistry questions: **effluents and pH, soaps, and an enthalpy reaction** (PYQ-DERIVED, Blueprint §4.4). They are §3, §4 and §5 of this chapter.

That is unusually precise guidance for one chapter, so use it. **Read §3, §4 and §5 properly.** Sections 1, 2 and 6 are recognition-level background — skim the tables and rely on the practice questions.

Chemistry sits between F9.1's pure recall and F9.5's genuine new learning. You will remember the structure; the *everyday-substance* facts are the part that needs actual attention.

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0. How This Chapter Works

STUDY SESSION · CHAPTER F9.3 — SIX SESSIONS

1. **Session 1** — Atomic structure and the mole (§1) ★★☆☆☆
2. **Session 2** — The periodic table and bonding (§2) ★★☆☆☆
3. **Session 3** — Acids, bases, salts and **pH** (§3) ★★★★★ **2023**
4. **Session 4** — **Soaps and detergents**, and everyday substances (§4) ★★★★★ **2023**
5. **Session 5** — **Enthalpy** and types of reaction (§5) ★★★★★ **2023**
6. **Session 6** — Metals, non-metals and carbon (§6) ★★★★★

Then 30 graded questions in §7. Budget **two and a half days**.

1. Session 1 — Atomic Structure and the Mole ★★☆☆☆

Particle	Charge	Location
Proton	Positive	Nucleus
Neutron	Neutral	Nucleus
Electron	Negative	Shells around the nucleus

- **Atomic number (Z) = number of protons.** It defines the element
- **Mass number (A) = protons + neutrons**
- **Valency** — the combining capacity, determined by the outermost shell and the **octet rule**

MUST MASTER · ISOTOPES, ISOBARS AND THE MOLE

Term	Same	Different
Isotopes	Atomic number (protons)	Mass number (neutrons)
Isobars	Mass number	Atomic number
Isotones	Number of neutrons	Both Z and A

Isotopes have the **same chemical** properties (same electron arrangement) but different **physical** properties. Examples: protium, deuterium and tritium; carbon-12 and carbon-14; uranium-235 and uranium-238.

One mole contains 6.022×10^{23} particles — Avogadro's number. That figure is asked directly, and its usual distractors are the speed of light (3×10^8) and the electronic charge (1.6×10^{-19}).

CHECK YOURSELF · SESSION 1

1. What does the atomic number count? What does the mass number count?
2. Distinguish isotopes from isobars. Which properties do isotopes share?
3. State Avogadro's number

2. Session 2 — Periodic Table and Bonding ★★☆☆☆

2.1 The periodic table

MUST MASTER · THE LAW, AND THE GROUP NAMES

The modern periodic law: the properties of elements are a periodic function of their ATOMIC NUMBER. Mendeleev's earlier law used **atomic mass** — that swap is the standard question.

There are **18 groups and 7 periods**. The group names that get asked:

Group	Name
1	Alkali metals
2	Alkaline earth metals
17	Halogens
18	Noble or inert gases

The commonest error is calling group 18 the halogens. **17 halogens, 18 noble gases.**

Trends. Across a period, left to right: atomic size **decreases**, electronegativity **increases**, metallic character **decreases**. Down a group: atomic size **increases**, metallic character **increases**, electronegativity **decreases**.

2.2 Bonding

Bond	Mechanism	Properties	Example
Ionic (electrovalent)	Transfer of electrons	High melting point; conducts when molten or in solution	NaCl
Covalent	Sharing of electrons	Low melting point; generally poor conductor	H ₂ O, CH ₄
Metallic	Delocalised electrons	Conducts, malleable, ductile	Iron, copper
Hydrogen bond	Weak attraction involving H and an electronegative atom	Raises boiling point	Why ice floats and water boils high

CHECK YOURSELF · SESSION 2

1. State the modern periodic law. What did Mendeleev's law use instead?
2. Name groups 1, 2, 17 and 18
3. What happens to atomic size across a period and down a group?
4. Ionic or covalent — which involves transfer, and which has the higher melting point?

3. Session 3 — Acids, Bases, Salts and pH ★★★★★

A 2023 topic. Read this section carefully.

3.1 Acids and bases

	Acid	Base
Taste	Sour	Bitter, and feels soapy
Litmus	Turns blue litmus red	Turns red litmus blue
In water	Releases H⁺ ions	Releases OH⁻ ions
Examples	HCl, H ₂ SO ₄ , citric, acetic	NaOH, KOH, Ca(OH) ₂

Neutralisation: acid + base » **salt + water**, and it is **exothermic**.

Indicators: litmus; methyl orange; **phenolphthalein** — **colourless in acid, PINK in base**; universal indicator for the whole range.

3.2 The pH scale

MUST MASTER · THE SCALE, AND THE FACTOR OF TEN

The **pH scale runs from 0 to 14**. $\text{pH} = -\log[\text{H}^+]$.

- **pH 7 — neutral** (pure water at 25 °C)
- **Below 7 — acidic**. The lower the number, the stronger the acid
- **Above 7 — basic or alkaline**

Each unit of pH is a factor of TEN in hydrogen ion concentration. So a solution of pH 4 has **100 times** the hydrogen ion concentration of one at pH 6 — two units, so 10^2 , not two times. That multiplication is a favourite numerical.

Values worth carrying:

Substance	pH
Gastric juice	about 1.5 to 3.5
Pure water	7
Human blood	about 7.4 — slightly alkaline , and tightly regulated
Milk	about 6.6

3.3 Effluents and pH — the 2023 item

MUST MASTER · THE DISCHARGE STANDARD

Under the general standards for the **discharge of industrial effluents into inland surface water**, the permissible **pH range is about 5.5 to 9.0**.

The logic is worth holding rather than the number alone: discharge is permitted across a **band either side of neutral**, not at exactly 7, because requiring precise neutrality would be unattainable — but strongly acidic or strongly alkaline effluent destroys aquatic life and corrodes infrastructure.

So the wrong options are "exactly 7.0", and any band lying wholly on one side of neutral such as 1 to 5 or 9 to 12. The correct answer **straddles 7**.

Related environmental chemistry, which connects to F9.4: **acid rain** is caused by oxides of sulphur and nitrogen forming sulphuric and nitric acids; **ocean acidification** is caused by dissolved carbon dioxide forming carbonic acid.

3.4 Salts and water hardness

Common name	Chemical name
Common salt	Sodium chloride, NaCl
Baking soda	Sodium hydrogen carbonate , NaHCO ₃
Washing soda	Sodium carbonate decahydrate
Bleaching powder	Calcium oxychloride
Quicklime	Calcium oxide, CaO
Slaked lime	Calcium hydroxide, Ca(OH) ₂
Plaster of Paris	Calcium sulphate hemihydrate
Caustic soda	Sodium hydroxide, NaOH

Hardness of water:

Type	Caused by	Removed by
Temporary	Bicarbonates of calcium and magnesium	Boiling
Permanent	Sulphates and chlorides of calcium and magnesium	Washing soda, ion exchange

CHECK YOURSELF · SESSION 3

1. Which way does each litmus paper turn for an acid and for a base?
2. Give the pH range, the neutral value, and the pH of blood
3. A solution of pH 3 versus one of pH 5 — how many times the hydrogen ion concentration?
4. What is the permissible pH band for effluent discharge into inland surface water?
5. Phenolphthalein — what colour in acid, and in base?
6. Temporary and permanent hardness — cause and cure of each

4. Session 4 — Soaps, Detergents and Everyday Substances ★★★★★

A 2023 topic.

4.1 Soaps and detergents

MUST MASTER · THE DEFINITIONS, AND THE HARD-WATER DIFFERENCE

Soap is the sodium or potassium salt of a long-chain fatty acid. It is made by **saponification** — the hydrolysis of fats or oils with an alkali such as sodium hydroxide.

A detergent is the sodium salt of a long-chain benzene sulphonic acid or sulphate.

Now the difference that generates the question:

	Soap	Detergent
Works in hard water ?	No — forms an insoluble scum	Yes
Why	Its calcium and magnesium salts are insoluble	Its calcium and magnesium salts are soluble
Biodegradability	Biodegradable	Often less biodegradable — branched chains cause foaming in rivers

Note the direction carefully, because reversing it is the standard wrong option: **it is SOAP that fails in hard water**, not the detergent. Detergents were developed precisely to solve that problem — and the environmental cost was reduced biodegradability.

How soap cleans — the micelle. A soap molecule has a **hydrophilic (water-loving) ionic head** and a **hydrophobic (water-repelling) hydrocarbon tail**. In water the molecules cluster into a **micelle** with the tails pointing inwards, trapping oil and grease in the centre while the heads face the water — so the dirt is carried away in suspension.

4.2 Everyday substances worth recognising

Name	What it is
Dry ice	Solid carbon dioxide
Laughing gas	Nitrous oxide, N_2O
Marsh gas	Methane, CH_4
Vinegar	Dilute ethanoic (acetic) acid , about 5 to 8%
Aqua regia	A mixture of concentrated HCl and concentrated HNO_3 , which dissolves gold
Producer gas, water gas	Fuel gas mixtures
Urea	A nitrogenous fertiliser

CHECK YOURSELF · SESSION 4

1. Define soap and detergent chemically. What is the soap-making process called?
2. Which of the two fails in hard water, and why?
3. Which is less biodegradable, and what is the environmental consequence?
4. Describe a micelle
5. Name: dry ice, laughing gas, marsh gas, aqua regia

5. Session 5 — Enthalpy and Types of Reaction ★★★★★

A 2023 topic.

5.1 Exothermic and endothermic

MUST MASTER · THE SIGN OF ΔH , AND THE TWO LISTS

	Exothermic	Endothermic
Heat	Released	Absorbed
ΔH	Negative	Positive
Surroundings	Warm up	Cool down

The sign is the trap. Heat coming *out* of the system makes the enthalpy change **negative**, which feels backwards to beginners — you are measuring the system's enthalpy, not the room's. An option saying an endothermic reaction has a negative ΔH is wrong.

Exothermic: combustion of fuels · **respiration** · **neutralisation** of an acid by a base · quicklime added to water · rusting (slow).

Endothermic: **photosynthesis** · **thermal decomposition**, such as calcium carbonate to calcium oxide and carbon dioxide · dissolving ammonium chloride in water · electrolysis.

The clean pair to remember: **respiration is exothermic, photosynthesis is endothermic**. They are the reverse of one another, which is why they carry opposite signs.

5.2 Types of reaction

Type	Description
Combination	Two or more reactants form one product
Decomposition	One reactant splits — thermal, electrolytic or photochemical
Displacement	A more reactive element displaces a less reactive one
Double displacement	Ions are exchanged; includes precipitation and neutralisation
Redox	Oxidation and reduction occur together

MUST MASTER · OIL RIG

Oxidation Is Loss, Reduction Is Gain — of ELECTRONS.

	Oxidation	Reduction
Electrons	Lost	Gained
Oxygen	Gained	Lost
Hydrogen	Lost	Gained

The modern definition is in terms of **electrons**, which is what "best defined as" questions are testing. The oxygen definition is older and narrower.

Corrosion: rusting of iron requires **both oxygen AND moisture** — either alone will not do, which is why an option naming only one is wrong. Prevented by painting, oiling, **galvanisation** (zinc coating) and alloying.

Rancidity: oxidation of fats and oils, retarded by **antioxidants** and by flushing packaging with **nitrogen**.

CHECK YOURSELF · SESSION 5

1. Which sign does ΔH take for an exothermic reaction?
2. Give three exothermic and three endothermic examples
3. Which of respiration and photosynthesis is endothermic?
4. Define oxidation in terms of electrons. What does OIL RIG stand for?
5. What two things does rusting require? Name three preventive methods

6. Session 6 — Metals, Non-metals and Carbon ★★★★★

6.1 Metals and non-metals

	Metals	Non-metals
Appearance	Lustrous	Dull, brittle
Conductivity	Good	Poor — except graphite
Oxides formed	Basic	Acidic
Exceptions	Mercury is liquid; sodium and potassium are soft	Bromine is liquid; graphite conducts; diamond is hardest

Reactivity series, most reactive first:



Alloys worth knowing:

Alloy	Composition
Brass	Copper + zinc
Bronze	Copper + tin
Steel	Iron + carbon
Stainless steel	Iron + chromium + nickel
Solder	Lead + tin
Duralumin	Aluminium + copper + magnesium + manganese
Amalgam	Any alloy containing mercury

6.2 Carbon

Carbon shows tetravalency and catenation — it forms four bonds and long chains, which is why organic chemistry exists.

Allotrope	Property
Diamond	Hardest natural substance; does not conduct electricity
Graphite	Conducts electricity; soft; used as a lubricant and in pencils
Fullerene	Cage-like molecules, such as C-60

Hydrocarbons: alkanes C_nH_{2n+2} (saturated) · alkenes C_nH_{2n} · alkynes C_nH_{2n-2} .

Functional groups: alcohol -OH · aldehyde -CHO · carboxylic acid -COOH.

COMMON UPSC TRAP · DIAMOND AND GRAPHITE

Both are pure carbon, and the exam relies on you knowing which property belongs to which.

Graphite conducts electricity; diamond does not. Graphite has delocalised electrons between its layers; diamond's electrons are all locked into covalent bonds. And **diamond is the hardest** natural substance while graphite is soft enough to write with.

Same element, opposite properties — because the difference is structure, not composition.

CHECK YOURSELF · SESSION 6

1. Do metals form basic or acidic oxides? And non-metals?
2. Name the liquid metal and the liquid non-metal
3. Recite the reactivity series from potassium to gold
4. Give the composition of brass, bronze, stainless steel and solder
5. Which allotrope of carbon conducts electricity, and which is hardest?

7. Graded Practice — 30 Questions ★★★★★

7.1 Level 1 — Recognition

- Q1.** The pH of a neutral solution at 25 °C is (a) 7 (b) 0 (c) 14 (d) 1
- Q2.** Which one of the following turns red litmus blue? (a) An acid (b) A neutral solution (c) A base (d) A solution of common salt
- Q3.** The atomic number of an element is equal to the number of (a) neutrons (b) protons (c) protons and neutrons together (d) electrons in the outermost shell
- Q4.** Isotopes of an element have (a) the same mass number but different atomic numbers (b) the same number of neutrons (c) different chemical properties (d) the same atomic number but different mass numbers
- Q5.** Avogadro's number is approximately (a) 6.022×10^{23} (b) 3.0×10^8 (c) 1.6×10^{-19} (d) 9.1×10^{-31}
- Q6.** Soap is chemically (a) a long-chain hydrocarbon (b) sodium benzene sulphonate (c) the sodium or potassium salt of a long-chain fatty acid (d) sodium carbonate
- Q7.** In an exothermic reaction the change in enthalpy is (a) positive (b) negative (c) zero (d) undefined
- Q8.** Which one of the following is an **endothermic** process? (a) Combustion of coal (b) Respiration (c) Neutralisation of an acid by a base (d) Photosynthesis
- Q9.** The chemical name of baking soda is (a) sodium hydrogen carbonate (b) sodium carbonate (c) calcium carbonate (d) sodium chloride
- Q10.** The hardest naturally occurring form of carbon is (a) graphite (b) fullerene (c) diamond (d) coke

7.2 Level 2 — Understanding

- Q11.** Soaps are ineffective in hard water because they (a) decompose in hard water (b) form an insoluble scum with calcium and magnesium ions (c) become acidic (d) evaporate rapidly
- Q12.** The process of making soap by hydrolysing fats or oils with an alkali is called (a) esterification (b) hydrogenation (c) polymerisation (d) saponification
- Q13.** Under the general standards for the discharge of industrial effluents into inland surface water, the permissible pH range is about (a) 5.5 to 9.0 (b) 1.0 to 5.0 (c) 9.0 to 12.0 (d) exactly 7.0
- Q14.** The pH of human blood is normally about (a) 5.5 (b) 7.0 (c) 7.4 (d) 2.0
- Q15.** Phenolphthalein in a basic solution is (a) colourless (b) pink (c) yellow (d) blue
- Q16.** In an ionic bond, electrons are (a) shared equally (b) shared unequally (c) delocalised over the whole lattice (d) transferred from one atom to another
- Q17.** The modern periodic law states that the properties of elements are a periodic function of their (a) atomic number (b) atomic mass (c) valency (d) density

- Q18.** Which gas is known as laughing gas? (a) Carbon dioxide (b) Methane (c) Nitrous oxide (d) Ammonia
- Q19.** The rusting of iron requires (a) oxygen only (b) both oxygen and moisture (c) moisture only (d) carbon dioxide only
- Q20.** Aqua regia is a mixture of (a) dilute hydrochloric and dilute nitric acid (b) concentrated sulphuric and concentrated nitric acid (c) hydrochloric and sulphuric acid (d) concentrated hydrochloric and concentrated nitric acid

7.3 Level 3 — Recruitment Test standard

Q21. Which one of the following statements is **not** correct? (a) Detergents, unlike soaps, form an insoluble scum with hard water (b) A soap molecule has a hydrophilic head and a hydrophobic tail (c) Detergents are generally less biodegradable than soaps (d) Soap is manufactured by the process of saponification

Q22. Consider the following statements:

1. Combustion is an exothermic reaction.
2. The thermal decomposition of calcium carbonate is endothermic.
3. In an endothermic reaction the enthalpy change is negative.

Which of the above statements are correct? (a) 1 and 3 only (b) 2 and 3 only (c) 1 and 2 only (d) 1, 2 and 3

Q23. Match List-I with List-II and select the correct answer using the code given below:

List-I (Alloy)	List-II (Composition)
A. Brass	1. Copper and zinc
B. Bronze	2. Copper and tin
C. Stainless steel	3. Iron, chromium and nickel
D. Solder	4. Lead and tin

(a) A-2 B-1 C-3 D-4 (b) A-1 B-2 C-3 D-4 (c) A-1 B-2 C-4 D-3 (d) A-3 B-4 C-1 D-2

Q24. A solution of pH 4, compared with a solution of pH 6, has a hydrogen ion concentration that is (a) 2 times greater (b) 2 times smaller (c) 100 times smaller (d) 100 times greater

Q25. Consider the following statements about the periodic table:

1. On moving from left to right across a period, atomic size generally decreases.
2. On moving down a group, metallic character generally increases.
3. The elements of group 18 are called the halogens.

Which of the above statements are correct? (a) 1 and 2 only (b) 2 and 3 only (c) 1 and 3 only (d) 1, 2 and 3

- Q26.** In the reactivity series, which metal is more reactive than iron but less reactive than magnesium? (a) Copper (b) Gold (c) Zinc (d) Silver
- Q27.** Temporary hardness of water is caused by, and removed by, respectively (a) sulphates; boiling (b) bicarbonates of calcium and magnesium; boiling (c) chlorides; filtration (d) nitrates; distillation
- Q28.** Which one of the following statements is **not** correct? (a) Graphite conducts electricity while diamond does not (b) Carbon exhibits tetravalency and catenation (c) Ionic compounds generally have high melting points (d) Non-metals generally form basic oxides
- Q29.** Oxidation is best defined as the (a) loss of electrons (b) gain of electrons (c) loss of oxygen (d) gain of hydrogen
- Q30.** Dry ice is (a) ice maintained at $-10\text{ }^{\circ}\text{C}$ (b) frozen water free of impurities (c) solid carbon dioxide (d) solid nitrogen

8. Answers and Explanations

8.1 Level 1

Q	Ans	Explanation
1	a	7. The scale runs 0 to 14, so 0 and 14 are the extremes, not neutral
2	c	A base turns red litmus blue. An acid turns blue litmus red — learn the pair together
3	b	The number of protons , which is what defines the element
4	d	Same atomic number, different mass numbers. Option (a) describes isobars , and (c) is false — isotopes share chemical properties
5	a	6.022×10^{23}. The distractors are the speed of light and the electronic charge
6	c	The sodium or potassium salt of a long-chain fatty acid . Option (b) describes a detergent
7	b	Negative — heat leaves the system, so its enthalpy falls
8	d	Photosynthesis absorbs energy. Combustion, respiration and neutralisation are all exothermic
9	a	Sodium hydrogen carbonate. Sodium <i>carbonate</i> is washing soda
10	c	Diamond

8.2 Level 2

Q	Ans	Explanation
11	b	They form an insoluble scum with the calcium and magnesium ions in hard water. Detergents do not, because their corresponding salts are soluble
12	d	Saponification — alkaline hydrolysis of fats and oils
13	a	5.5 to 9.0 — a band straddling neutral. "Exactly 7.0" is unattainable in practice, and bands lying wholly on one side are wrong
14	c	About 7.4 , so blood is slightly alkaline and is tightly regulated
15	b	Pink in base, colourless in acid
16	d	Transferred. Sharing is covalent; delocalisation is metallic
17	a	Atomic number. Mendeleev's earlier law used atomic mass , which is option (b) and the standard distractor
18	c	Nitrous oxide. Marsh gas is methane
19	b	Both oxygen and moisture. Either alone is insufficient, which is why the single-factor options are wrong
20	d	Concentrated HCl and concentrated HNO ₃ — the concentration matters, so (a) is wrong even though the acids are right

8.3 Level 3

Q	Ans	Explanation
21	a	Reversed. It is soap that forms scum in hard water; detergents do not. The other three statements are correct
22	c	1 and 2 only. Statement 3 inverts the sign — an endothermic reaction has a positive ΔH
23	b	A-1 B-2 C-3 D-4. Brass is copper and zinc; bronze copper and tin. These two are the pair most often swapped
24	d	Each pH unit is a factor of ten , so two units is $10^2 = 100$ times greater . Lower pH means more acidic, hence more hydrogen ions. Option (a) treats the difference as arithmetic rather than logarithmic
25	a	1 and 2 only. Statement 3 is false — group 17 are the halogens; group 18 are the noble gases
26	c	Zinc. The series runs $K > Na > Ca > Mg > Al > Zn > Fe$, so zinc sits between magnesium and iron
27	b	Bicarbonates , removed by boiling . Permanent hardness comes from sulphates and chlorides and needs washing soda or ion exchange
28	d	Non-metals form acidic oxides, not basic. Metals form basic oxides. The other three statements are correct
29	a	Loss of electrons — the modern definition, and what "best defined" is asking for. The oxygen-based definition is older and narrower
30	c	Solid carbon dioxide , which sublimates without melting

THINK LIKE UPSC · THE FOUR SHAPES IN THIS CHAPTER

- The reversal.** Q21 and Q28 both work by swapping which of a pair has a property — soap versus detergent, metal versus non-metal. Whenever two things are contrasted, check the direction
- The logarithm.** Q24. pH differences are **powers of ten**, never arithmetic. Two units means 100 times
- The sign of ΔH .** Q7 and Q22. Exothermic is **negative**, which feels backwards until you remember you are measuring the system, not the room
- The concentration qualifier.** Q20 offers the right acids at the wrong concentration. Read the adjectives in chemistry options, not just the formulas

9. One-Minute Revision

ONE-MINUTE REVISION · CHAPTER F9.3

ATOM protons + neutrons in the nucleus, electrons outside · **Z = protons** · **A = protons + neutrons** · **isotopes: same Z, different A**, same chemical properties · **isobars: same A** · Avogadro 6.022×10^{23}

PERIODIC LAW properties are a periodic function of **ATOMIC NUMBER** (Mendeleev used mass) · **18 groups, 7 periods** · **1 alkali** · **2 alkaline earth** · **17 HALOGENS** · **18 NOBLE GASES** · across a period size **decreases** · down a group metallic character **increases**

BONDS ionic = **transfer**, high melting point, conducts when molten · covalent = **sharing** · metallic = delocalised · hydrogen bond explains why **ice floats**

ACIDS AND BASES acid turns **blue litmus RED**, releases H^+ · base turns **red litmus BLUE**, releases OH^- · **phenolphthalein: colourless in acid, PINK in base** · neutralisation is **exothermic**

pH scale **0 to 14** · **7 neutral** · each unit is a factor of **TEN** · pH 4 versus pH 6 = **100 times** more H^+ · **blood about 7.4, slightly alkaline** · gastric juice 1.5–3.5

EFFLUENT DISCHARGE pH 5.5 TO 9.0 — a band **straddling** neutral · acid rain from oxides of sulphur and nitrogen · ocean acidification from dissolved **CO₂**

HARDNESS temporary = **bicarbonates**, removed by **boiling** · permanent = **sulphates and chlorides**, removed by washing soda or ion exchange

SOAP = sodium or potassium salt of a **long-chain fatty acid** · made by **SAPONIFICATION** · **FAILS in hard water**, forms insoluble **scum** · **biodegradable DETERGENT** = sodium salt of a long-chain **benzene sulphonic acid** · **works in hard water** · **less biodegradable**, causes foaming **MICELLE** hydrophilic **head** out, hydrophobic **tail** in, trapping oil

ENTHALPY exothermic = heat **released**, **ΔH NEGATIVE** — combustion, **respiration**, neutralisation, quicklime in water · endothermic = heat **absorbed**, **ΔH POSITIVE** — **photosynthesis**, thermal decomposition, electrolysis

OIL RIG Oxidation Is Loss, Reduction Is Gain — of **electrons**

RUSTING NEEDS OXYGEN AND MOISTURE, BOTH · prevented by painting, oiling, **galvanisation** (zinc), alloying · **rancidity** retarded by antioxidants and nitrogen flushing

METALS basic oxides, good conductors, **mercury liquid** **NON-METALS** acidic oxides, poor conductors **except graphite, bromine** liquid

REACTIVITY $K > Na > Ca > Mg > Al > Zn > Fe > Pb > (H) > Cu > Hg > Ag > Au$

ALLOYS brass = Cu + **Zn** · bronze = Cu + **Sn** · stainless steel = Fe + **Cr + Ni** · solder = Pb + Sn · amalgam contains **mercury**

CARBON tetravalency and catenation · **diamond hardest, non-conductor** · **graphite conducts**, soft, lubricant · fullerene C-60

NAMES dry ice = **solid CO₂** · laughing gas = **N₂O** · marsh gas = **CH₄** · aqua regia = **concentrated HCl + concentrated HNO₃** · vinegar = dilute ethanoic acid

10. Five-Minute Wall Sheet

FIVE-MINUTE WALL SHEET · CHAPTER F9.3

pH 0–14 · 7 NEUTRAL · EACH UNIT = $\times 10$ · BLOOD 7.4 ALKALINE

EFFLUENT pH 5.5–9.0 — straddles neutral

ACID » BLUE TO RED · BASE » RED TO BLUE · PHENOLPHTHALEIN PINK IN BASE

SOAP FAILS IN HARD WATER (SCUM) · DETERGENT WORKS BUT IS LESS BIODEGRADABLE

SAPONIFICATION MAKES SOAP · MICELLE: TAIL IN, HEAD OUT

EXOTHERMIC = ΔH NEGATIVE · ENDOTHERMIC = ΔH POSITIVE

RESPIRATION EXOTHERMIC · PHOTOSYNTHESIS ENDOTHERMIC

OIL RIG — LOSS AND GAIN OF ELECTRONS

RUSTING = OXYGEN + MOISTURE, BOTH

GROUP 17 HALOGENS · GROUP 18 NOBLE GASES

PERIODIC LAW USES ATOMIC NUMBER (Mendeleev used mass)

IONIC = TRANSFER · COVALENT = SHARING

METALS BASIC OXIDES · NON-METALS ACIDIC OXIDES

BRASS = Cu + Zn · BRONZE = Cu + Sn

GRAPHITE CONDUCTS · DIAMOND HARDEST

TEMPORARY HARDNESS » BOIL IT

DRY ICE = SOLID CO_2 · LAUGHING GAS = N_2O · MARSH GAS = CH_4

ONE LINE FOR THE INTERVIEW Detergents were invented to solve a real problem — soap forms an insoluble scum in hard water — and created a new one, because their branched chains resist biodegradation and foam in rivers, which is a good small illustration of why environmental regulation follows technology rather than preceding it.

ACTION · NEXT IN MODULE 9

Chapter F9.4 — Biology and Environment, which covers the last three 2023 science items: the **layers of the eyeball**, **banyan prop roots**, and **ocean acidification**.

Before you move on, state without looking: the pH band for effluent discharge; which of soap and detergent fails in hard water; the sign of ΔH for an exothermic reaction; what OIL RIG stands for; and the compositions of brass and bronze.